Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Fe-Induced Autosomal Mutations in Murine Tissue and Cell Lines		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/01/2003	**End Date:**	04/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Turker, Mitchell (Oregon Health and Sciences University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Connolly, Lanelle (Oregon Health and Sciences University)		
Grant/Contract No.:	T-403-X		
Performance Goal No.:			
Performance Goal Text:			
	Cancer induction in humans involves a variety of genetic and epigenetic alterations, including widespread loss of expression of genes located primarily on autosomes. Currently, no in vivo data are available to determine the risks of mutagenesis induced by space radiations at an autosomal locus. This study will determine the mutagenic risks of Fe ions using a model autosomal locus (Aprt) in the B6D2 mouse. Four aims are proposed: Aim 1 tests the hypothesis that Fe ions induce autosomal mutations at the Aprt locus in B6D2 kidney epithelial cells exposed in vivo. Aim 2 tests the hypothesis that the exposure to Fe ions elicits primarily multi-locus deletions inclusive of Aprt in B6D2 kidney epithelial cells exposed in vivo. Aim 3 compares Fe-induced mutant frequencies in B6D2 kidney epithelial cells exposed in vivo or in vitro to test the hypothesis that the primary determinant of mutagenic risk is the energy deposited in the cells and not potential modulating effects of the cellular microenvironment. Aim 4 tests the hypothesis that the primary determinant of		

Task Description:	mutational spectrum is the energy deposited in the cells at risk by comparing the spectrum of Aprt mutants arising after in vitro exposure of B6D2 kidney epithelial cell lines to Fe ions with the spectrum of Aprt mutants obtained after in vivo exposure of kidney epithelial cells (Aim 2). Most literature on the biological effects of heavy ions describes results obtained with in vitro assays, yet considerable uncertainty exists as to how well these assays predict in vivo responses, particularly for endpoints relevant to carcinogenesis. Upon completion of the proposed work, NASA should be in a better position to determine whether cellular assays can be relied upon with a high degree of confidence, or whether greater emphasis should be placed on the establishment and testing of animal models. The goal of this study is to determine whether the tissue microenvironment modifies the mutagenic response of epithelial cells to charged particle radiation of importance in spaceflight. Mutant frequencies and mutant spectra will be evaluated in isogenic kidney epithelial cells exposed either in vitro or in vivo. Little information is available on the mutagenic effects of charged particle radiations in epithelial cell systems. The primary goal of this work is to establish such data and to determine how or if the cellular microenvironment modulates the expression of mutagenic damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will be useful for the assessment of the importance of the cellular microenvironment on the frequency and types of mutations induced at autosomal loci following exposure to ionizing radiation, with a particular emphasis on densely ionizing radiations such as alpha particles. These studies may also be of interest to the medical community as it considers the use of charged particles more widely in radiotherapy applications.
Task Progress:	Progress this year has gone according to plans. In vitro experiments at NSRL in late Spring 2003 and Fall 2003 provided the first data on the cytoxic effects of densely ionizing Fe ions (1 GeV/amu) on mouse kidney epithelial cells derived from two different strains of mice, and comparisons were made with sparsely ionizing X-rays as a reference radiation. In each case, Fe ions were more cytotoxic to kidney epithelial cells than X-rays per unit dose. New experiments in Spring 2004 provided the first quantitative data on autosomal mutagenesis in an epithelial cell type from a mammalian organism exposed to densely ionizing radiation. A dose-response was observed over the range of 0-200 cGy for mutation induction at the Aprt locus in kidney cells from B6D2 mice. The initial in vivo comparison experiments were also performed in Spring 2004, and data analysis is underway with collection times from 2-4 months post-irradiation. Individual mutant colonies from control cells or kidney epithelial cells exposed to 2 Gy of Fe ions are being developed for molecular characterization of mutant spectra. These studies are underway.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg, A.; Gauny, S.S.; Connolly, L.; Turker, M. "Genotoxic effects of 1 GeV/amu Fe ions in mouse kidney epithelial cells." 51st Annual Meeting of the Radiation Research Society, St. Louis, MO. N/A , Apr-2004
Abstracts for Journals and Proceedings	Kronenberg, A.; Gauny, S.S.; Connolly, L.; Turker, M. "Genotoxic effects of 1 GeV/amu Fe ions in mouse kidney epithelial cells." Third International Workshop on Space Radiation Research, Port Jefferson, NY. page 13 , May-2004
Abstracts for Journals and Proceedings	Durante, M.; Kronenberg, A. "Ground-based iron-ion biology and physics (introduction)." 35th International Congress on Space Research (COSPAR), Paris, France. F 2.3-0001-04 , Jul-2004
Abstracts for Journals and Proceedings	Kronenberg, A.; Gauny, S.S.; Connolly, L.; Turker, M. "Genotoxic effects of 1 GeV/amu Fe ions in mouse kidney epithelial cells." 35th COSPAR Scientific Assembly, Paris France. F2.3-0016-04 , Jan-2004